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Major Project Report
MACHINE LEARNING CONCEPT – 2 (CSE 3968)

On

OncoVision AI

Bridging the Gap Between AI Diagnostic Power and Clinical Understanding

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1 Abstract

OncoVision AI is a high-precision medical diagnostic platform designed to assist clinicians and pathologists in the detection of breast cancer.

The system integrates Explainable Artificial Intelligence (XAI), machine learning, and multimodal pathology analysis to provide transparent and clinically reliable predictions.

Unlike traditional black-box diagnostic systems, OncoVision AI uses SHAP (SHapley Additive Explanations) to generate feature-level interpretability for every prediction.

The platform also integrates a Gemini-powered Intelligent Report Gateway capable of extracting cytological information directly from pathology reports.

The core objective of the project is to minimize False Negatives in malignant cancer detection while improving clinical trust, interpretability, and accessibility.

2 Introduction

Artificial Intelligence is rapidly transforming modern healthcare systems by enabling automated diagnostics, predictive analytics, and intelligent decision support.

Breast cancer remains one of the most common and life-threatening diseases affecting women worldwide. Early-stage malignant detection is critical because delayed diagnosis can significantly reduce survival probability.

Traditional AI diagnostic systems often function as black-box models, making it difficult for clinicians to understand the reasoning behind predictions.

OncoVision AI was developed to address these challenges by combining:

- Recall-focused machine learning
- Explainable AI
- Multimodal pathology analysis
- Patient-friendly diagnostic workflows

3 Problem Statement

Current AI diagnostic systems face multiple clinical limitations:

- Many systems prioritize overall accuracy rather than malignant detection sensitivity.
- False Negative predictions can delay life-saving treatment.
- Black-box AI models reduce clinician trust.

- Manual pathology interpretation requires significant technical expertise.
- Existing systems often lack patient-friendly accessibility.

The objective of this project is to develop a transparent and clinically reliable diagnostic platform capable of minimizing False Negatives while improving explainability.

4 Objectives

The primary objectives of OncoVision AI are:

- Develop a Recall-First breast cancer diagnostic engine
- Integrate Explainable AI using SHAP
- Automate pathology report analysis using Gemini OCR
- Improve clinician trust through transparent predictions
- Create a scalable full-stack clinical AI platform

5 Literature Review

Machine learning applications in healthcare have significantly improved disease prediction and clinical decision support systems.

Several existing oncology AI systems utilize algorithms such as:

- Artificial Neural Networks
- Random Forest
- Support Vector Machines
- Deep Learning architectures

However, many healthcare AI systems suffer from:

- Limited interpretability
- Black-box predictions
- Poor clinical transparency
- Insufficient False Negative optimization

Recent Explainable AI techniques such as SHAP have emerged as important tools for improving clinician trust and understanding prediction behavior.

Research also shows that Recall-focused optimization is critical in oncology because missed malignant cases are clinically dangerous.

6 Technical Foundation: Dataset Specifications

The diagnostic engine was trained using the Breast Cancer Wisconsin (Diagnostic) Dataset.

6.1 Dataset Overview

- Total Samples: 569
- Feature Count: 30
- Benign Cases: 357 (62.7%)
- Malignant Cases: 212 (37.3%)

6.2 Key Cytological Features

- Radius
- Texture
- Perimeter
- Area
- Smoothness
- Compactness
- Concavity
- Concave Points

These features were extracted from digitized Fine Needle Aspiration images of breast masses.

7 Methodology

7.1 Data Preprocessing

The dataset was preprocessed using:

- StandardScaler normalization
- Feature extraction
- Train-test splitting
- Data validation

7.2 Model Development

Multiple machine learning algorithms were evaluated:

- Artificial Neural Network (ANN)
- Random Forest
- Support Vector Machine (SVM)

7.3 Model Evaluation

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC
- Confusion Matrix

8 The Diagnostic Engine: Model Selection

Following comparative experimentation, the Support Vector Machine (SVM) classifier was selected as the final diagnostic engine.

8.1 Comparative Performance

Model	**Technique**	**Observation**
ANN	Deep Learning	High accuracy but prone to overfitting
Random Forest	Ensemble Learning	Lower recall performance
SVM	Margin Maximization	Highest Recall and stable clinical performance

Table 1: Comparative evaluation of machine learning models

8.2 Why SVM?

1. Clinical Safety through Recall optimization

2. RBF Kernel support for non-linear feature relationships
3. Stable SHAP interpretability integration
4. Strong performance on high-dimensional medical data

9 Final Performance Metrics

Metric	Value	Clinical Significance
Accuracy	96.49%	Overall correctness
Recall	97.62%	Minimized False Negatives
Precision	93.18%	Reliable malignant prediction confidence
F1 Score	95.35%	Balanced precision and recall
ROC-AUC	0.9957	Excellent class discrimination

Table 2: Final diagnostic performance metrics

10 Performance Validation: Confusion Matrix

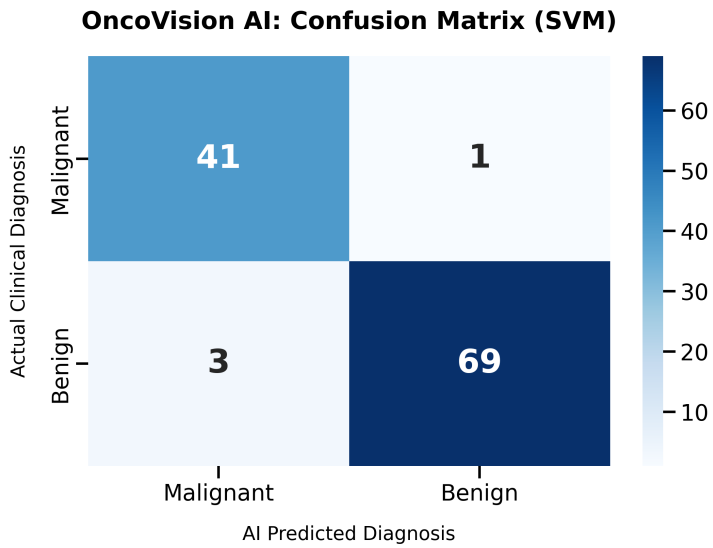

Figure 1: SVM Confusion Matrix

The confusion matrix demonstrates that only 1 malignant sample was incorrectly classified as benign.

This validates the Recall-First diagnostic philosophy adopted by the system.

11 Explainable AI: SHAP Implementation

To address black-box AI limitations, OncoVision AI integrates SHAP-based interpretability.

11.1 Implementation Strategy

- KernelExplainer integration
- Global feature importance visualization
- Patient-level explanations
- Clinician-verifiable prediction transparency

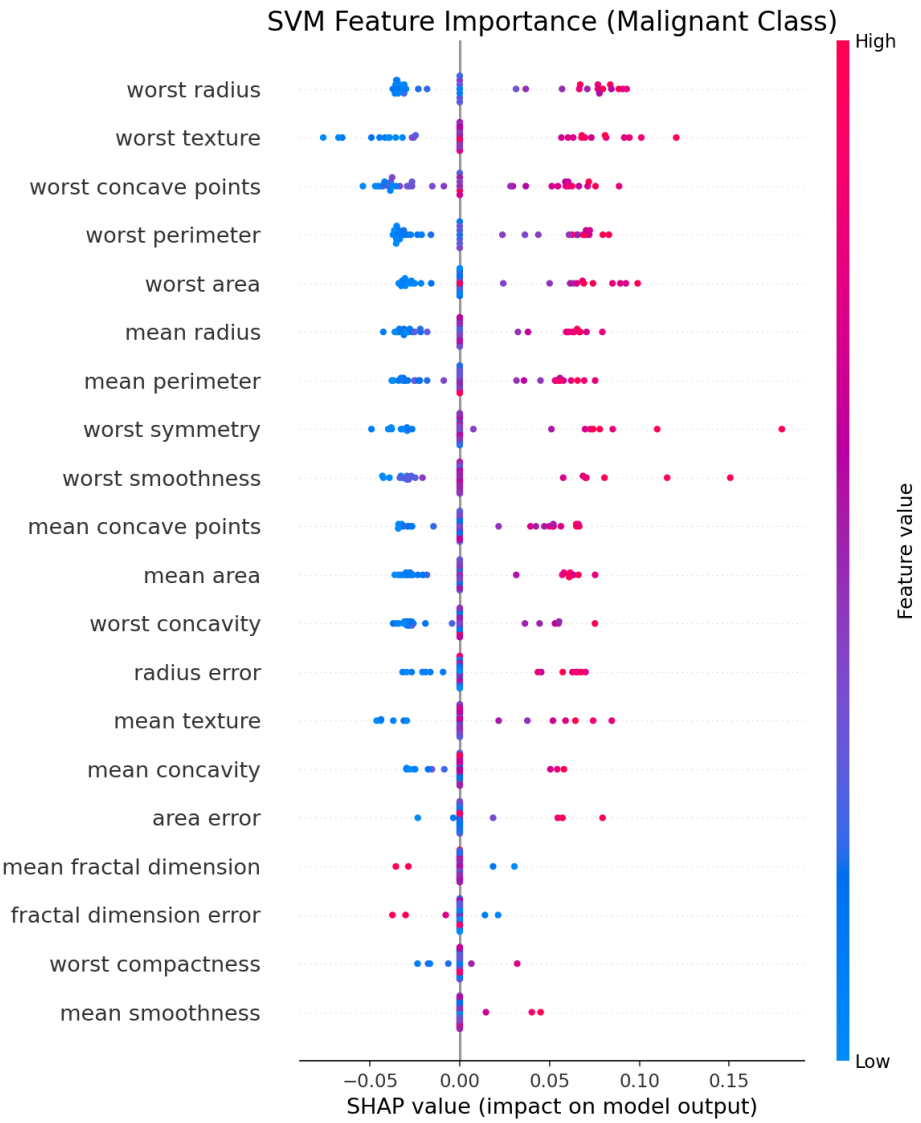

Figure 2: SHAP Summary Plot

11.2 Clinical Value

The SHAP module allows clinicians to understand which features contributed to malignant predictions.

Features such as:

- Concave Points
- Perimeter Worst
- Radius

emerged as dominant malignant indicators.

12 System Architecture

The platform follows a modular clinical AI architecture.

Figure 3: Clinical AI Infrastructure

The architecture includes:

- Frontend interaction layer

- FastAPI backend inference layer
- Gemini OCR intelligence layer
- SVM prediction engine
- SHAP explainability module

13 Intelligent Report Gateway Workflow

The Gemini-powered Intelligent Report Gateway automates pathology report analysis.

13.1 Workflow Pipeline

1. Upload pathology report
2. Gemini multimodal OCR processing
3. Clinical relevance validation
4. Cytological feature extraction
5. Structured JSON generation
6. Auto-population into SVM engine

13.2 System Advantage

The gateway significantly reduces technical barriers in pathology interpretation workflows.

14 Technical Implementation

14.1 Technology Stack

Category	**Technology**
Machine Learning	Scikit-Learn, SVM, SHAP
Generative AI	Google Gemini 2.5 Flash
Backend	FastAPI, Python 3.12
Frontend	Next.js 15, TypeScript
Deployment	Vercel, Render, Cloudflare Tunnel

Table 3: Technology stack used in OncoVision AI

15 Results

The final system successfully achieved:

- High Recall malignant detection
- Explainable AI transparency
- Automated pathology report understanding
- Full-stack clinical AI deployment
- SHAP-based prediction interpretability

The project demonstrates how Explainable AI can improve trust and reliability in healthcare diagnostics.

16 Discussion

16.1 Strengths

- Recall-focused clinical optimization
- Explainable AI integration
- Scalable architecture
- Transparent prediction workflow
- Automated multimodal report processing

16.2 Limitations

- Limited dataset size
- Requires broader clinical validation
- OCR performance depends on report quality
- Not intended for autonomous diagnosis

17 Applications

Potential applications of OncoVision AI include:

- Clinical decision support systems
- Hospital diagnostic assistance
- Telemedicine integration
- Pathology workflow automation
- Healthcare AI research

18 Future Scope

Future improvements planned for the platform include:

- DICOM-based mammography integration
- Hospital-scale batch screening
- Ensemble learning architectures
- Mobile healthcare monitoring
- HIPAA-compliant deployment pipelines

19 Conclusion

OncoVision AI demonstrates how Explainable AI and Recall-focused machine learning can improve trust in clinical diagnostics.

By integrating:

- SVM-based classification
- SHAP interpretability
- Gemini-powered pathology analysis
- Full-stack deployment infrastructure

the system creates a transparent and clinically reliable healthcare AI ecosystem.

The platform prioritizes:

- Clinical safety

- Explainability
- Accessibility
- Workflow automation

making it a strong foundation for future intelligent oncology support systems.

20 Appendix

20.1 Dataset Statistics

Feature	**Mean Value**	**Standard Deviation**
Radius	14.13	3.52
Texture	19.29	4.30
Area	654.89	351.91
Concave Points	0.048	0.038

Table 4: Statistical overview of selected dataset features

Clinical Disclaimer:

OncoVision AI is a diagnostic support tool designed to assist healthcare professionals. It is not a replacement for clinical judgment and should be used only as a secondary verification system.

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